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BILINGUAL ARABIC–ENGLISH SPEECH-ACTIVATED COMMAND RECOGNITION SYSTEM

Part One of the Graduation Project Report

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CHAPTER ONE: INTRODUCTION

Introduction 1.1

Speech is a natural and increasingly important interface for interacting with computers and digital applications. A speech-activated command system allows a user to issue a short spoken instruction instead of relying exclusively on a keyboard, mouse, or touch interface. For a predefined command vocabulary, Keyword Spotting (KWS) is a suitable formulation: the system detects whether a short audio segment corresponds to one of the supported commands and can reject speech that does not belong to the command set. Unlike general-purpose Automatic Speech Recognition (ASR), KWS does not aim to transcribe arbitrary sentences.

This project studies a lightweight command-recognition system for Arabic and English. It uses the Arabic Speech Commands dataset and Google Speech Commands as the principal data sources. The proposed investigation compares language-specific models with a bilingual model, considers differences between the datasets, and emphasizes evaluation on speakers excluded from training. The system is intended to map an accepted command label to a restricted application action; it is not a translation system and does not claim to understand unrestricted speech.

Scientific Problem 1.2

The scientific problem is to determine whether lightweight keyword-spotting models can recognize predefined commands in Arabic and English while generalizing to speakers and acoustic conditions not observed during training. The problem is not limited to classification accuracy on a familiar test set. A model may perform well on recordings similar to its training data but degrade when the speaker, microphone, background noise, accent, or recording environment changes.

The two source datasets differ in size, speaker diversity, recording conditions, and vocabulary. The Arabic Speech Commands corpus is described as containing 12,000 one-second recordings across 40 Modern Standard Arabic command classes from 30 speakers. Google Speech Commands v0.02 is commonly described as containing approximately 105,829 recordings across 35 English words; the exact downloaded release must be verified before experiments. Only 19 lexical matches are identified between the vocabularies. These differences create risks of class imbalance, domain mismatch, and negative transfer when the datasets are combined (Ghandoura et al., 2021; Warden, 2018).

Accordingly, the central research question is: How do lightweight KWS architectures generalize to unseen speakers in Arabic and English, and how do bilingual joint training and controlled data augmentation affect that generalization? The answer must be established experimentally; no performance outcome is assumed in this report.

Project Aim and Objectives 1.3

The main aim is to design and evaluate a reproducible speech-activated command-recognition pipeline for a predefined Arabic–English vocabulary, with explicit rejection of unknown speech and silence.

The specific objectives are:

- Establish Arabic-only and English-only baseline models using consistent audio preprocessing and documented data splits.
- Compare lightweight model families, primarily Depthwise Separable CNN (DS-CNN) and Convolutional Recurrent Neural Network (CRNN), with Keyword Transformer (KWT-1) as an optional comparison when resources permit.
- Build a bilingual shared-label experiment using the 19 identified lexical matches, while preserving language-specific labels where no literal match exists.
- Evaluate generalization using speaker-disjoint test data and, if feasible, an additional challenge set recorded from new speakers.
- Investigate the effect of training-only augmentation, including background-noise mixing and small temporal shifts, and compare Log-Mel features with MFCCs as a controlled feature ablation.
- Measure classification quality, false acceptance and rejection, and model efficiency without presenting unmeasured values as findings.
- Map accepted command labels to a restricted set of application actions, with confidence-thresholding and optional temporal checks to reduce unintended triggers.

State of the Art 1.4

Research on small-footprint keyword spotting has explored convolutional architectures designed to reduce computational cost. Sainath and Parada (2015) demonstrated the use of convolutional neural networks for limited-vocabulary recognition, while Zhang et al. (2017), in “Hello Edge,” focused on keyword spotting under microcontroller constraints. These works

motivate measuring both recognition performance and resource requirements rather than optimizing accuracy alone.

The Keyword Transformer (KWT) applies self-attention to keyword spotting and provides a transformer-based comparison to convolutional approaches (Berg et al., 2021). In the benchmark summary included in the project's reference materials, KWT variants performed strongly on Google Speech Commands and a newly collected speaker extension. The same summary reported a substantial generalization drop for a plain DNN and highlighted the value of evaluation on newly recorded speakers. Those results are evidence from the reported benchmark setting, not a guarantee that the same model ranking will hold for Arabic commands or for this project's implementation.

The Arabic Speech Commands dataset provides a benchmark specifically for Arabic small-footprint keyword spotting (Ghandoura et al., 2021). Google Speech Commands provides a widely used English dataset for limited-vocabulary speech recognition (Warden, 2018). Their combination enables a controlled bilingual investigation, but differences in corpus size and recording domain must be treated as experimental factors rather than ignored. The project methodology therefore recommends reporting results separately by language before presenting pooled bilingual results.

Feature representation is another important design choice. Mel-frequency representations approximate the frequency resolution of human hearing and are commonly used for compact speech models. Davis and Mermelstein (1980) established a foundational comparison of parametric representations including MFCCs. The proposed system uses 40-bin Log-Mel filterbank features as the primary representation and MFCCs as a secondary comparison. Training-only augmentation, including time/frequency masking as introduced by SpecAugment, may be tested to improve robustness (Park et al., 2019).

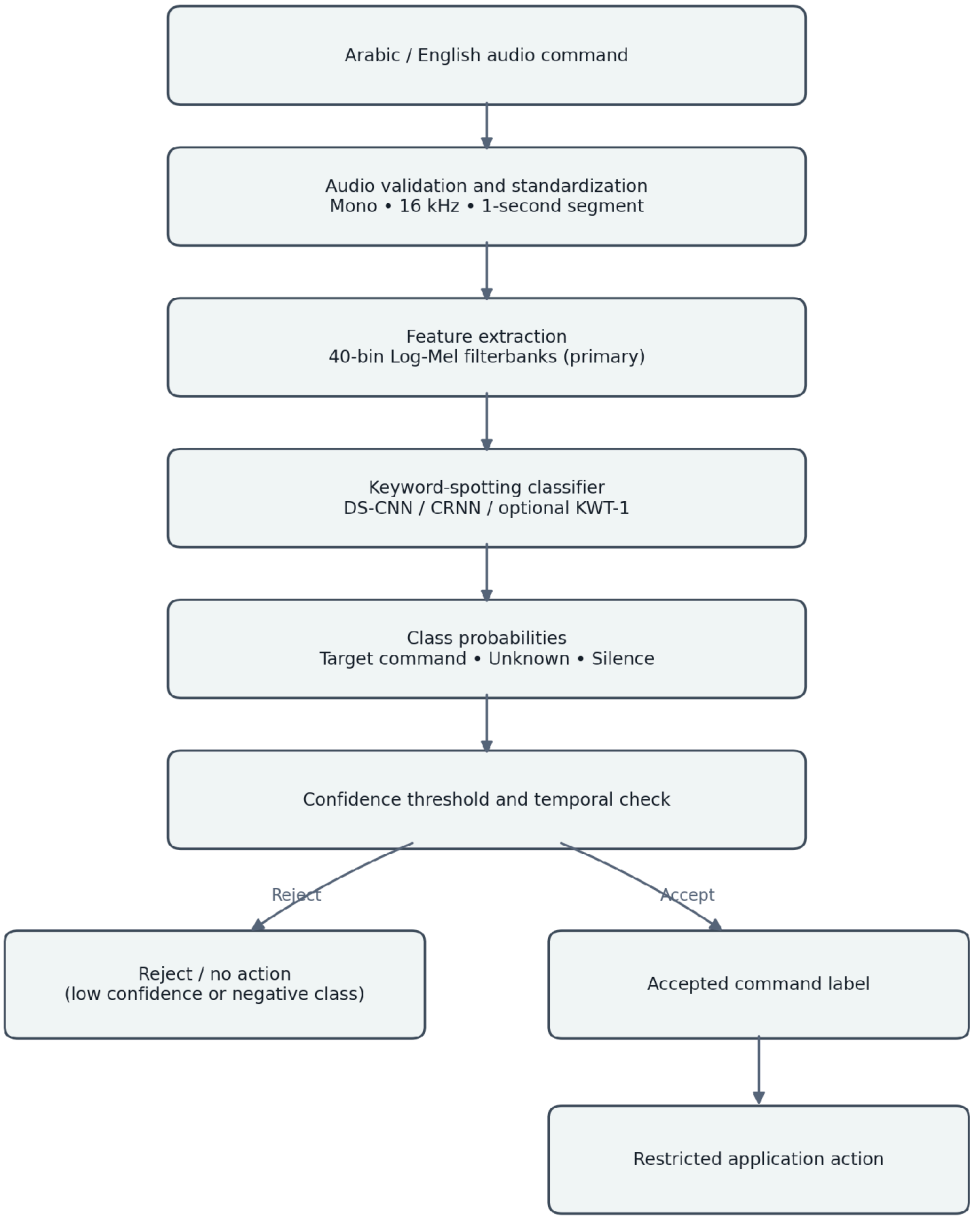
The project's research gap is therefore practical and comparative: the available references establish useful architectures, feature methods, and datasets, but they do not determine in advance how these choices will behave in a bilingual Arabic–English command system under speaker-disjoint evaluation. This project addresses that gap through a reproducible comparison and explicit analysis of language balance, unknown/silence rejection, and efficiency.

Proposed Solution and System Overview 1.5

The proposed solution consists of a preprocessing stage, acoustic feature extraction, a keyword-spotting classifier, and a command-decision stage. Audio

is converted to mono, resampled to 16 kHz when needed, and padded or cropped to a one-second segment. The primary feature representation is a 40-bin Log-Mel filterbank; MFCCs are retained for a controlled comparison. The classifier predicts a distribution over supported command labels and negative classes representing unknown speech and silence. A confidence threshold selected on validation data determines whether the predicted command is accepted. An accepted label is then mapped to an allowlisted application action.

Two bilingual operating designs are possible: a language-conditioned mode in which Arabic or English is selected by the user/application, or a shared bilingual classifier using a unified label inventory. The primary shared-vocabulary experiment uses the 19 literal lexical matches documented in the dataset analysis. Semantically related but phonetically different words are not merged as if they were identical labels.



.Figure 1.1. Proposed speech-command recognition and decision pipeline

The decision stage must reject low-confidence predictions and negative classes rather than trigger an action. In a streaming implementation, consecutive-frame agreement, hysteresis, or a cooldown interval may be evaluated to reduce repeated activations. These are design options to be tested, .not completed features or measured results

Expected Research Contribution 1.6

The intended contribution is a controlled, reproducible comparison of language-specific and bilingual keyword-spotting systems for a predefined Arabic–English command vocabulary. The study will document label alignment, protect speaker-disjoint evaluation, compare compact architectures and feature representations, and examine the trade-off between recognition performance and inference cost. The contribution lies in the experimental protocol and evidence produced by the implemented system; no new algorithmic contribution or numerical performance gain is claimed in advance.

Beneficiaries and Potential Use 1.7

Potential beneficiaries include researchers and students working on speech interfaces, developers of lightweight voice-controlled applications, and users who may benefit from an alternative input modality. The system may support simple predefined actions in desktop or embedded applications. Its scope is limited to the command vocabulary and acoustic conditions that are evaluated; it should not be treated as a general-purpose speech recognizer or as a safety-critical controller without additional validation.

Organization of This Report Part 1.8

This first part contains two chapters. Chapter One introduces the scientific problem, objectives, state of the art, and proposed system. Chapter Two presents the theoretical concepts required to understand the implementation, including keyword spotting, audio preprocessing, feature extraction, model architectures, bilingual label design, generalization, and evaluation metrics. Detailed implementation procedures, experimental results, and conclusions belong to later parts of the complete report and are not included here.

CHAPTER TWO: THEORETICAL STUDY

Keyword Spotting and Automatic Speech Recognition 2.1

Keyword Spotting (KWS) is a classification task in which an audio segment is assigned to a member of a predefined set of words or commands. A practical KWS system also needs to handle segments that contain no target command. In this project, the output space therefore includes supported command labels and negative classes such as unknown speech and silence. This distinction is important because a system that always selects a command may activate an application in response to irrelevant speech or background noise.

Automatic Speech Recognition (ASR), by contrast, aims to transcribe spoken language into a word sequence and can support open-ended utterances. The proposed project does not implement general ASR, translation, or natural-language understanding. It recognizes predefined labels and passes accepted labels to an action-mapping component. This narrower scope makes small-footprint models and explicit rejection decisions appropriate to investigate (Warden, 2018; Sainath & Parada, 2015).

Audio Signals and Standardization 2.2

A digital audio signal is a sequence of amplitude samples. The sampling rate specifies how many samples are stored per second. A rate of 16 kHz means 16,000 samples per second. A one-second mono segment at this rate contains 16,000 samples. Standardizing sample rate, channel count, and duration makes input dimensions consistent across examples and model architectures.

The proposed preprocessing converts each file to mono, resamples it to 16 kHz if necessary, and pads or crops it to one second using a documented policy. Signal checks should include file readability, duration, clipping, peak amplitude, silence, and channel count. Unusual files should be excluded only according to predefined criteria, and exclusions should be logged. Silence should not be removed indiscriminately because silence and segment boundaries may be relevant to negative-class detection and streaming use.

Time–Frequency Representations 2.3

Speech contains information that changes over time and frequency. A short-time Fourier transform (STFT) analyzes overlapping windows of the waveform to represent how spectral energy changes across frames. A window function, such as the Hann window, reduces edge discontinuities. The resulting

magnitude or power spectrum can be passed through a Mel filterbank, whose frequency bands are spaced to reflect a perceptually motivated frequency scale.

A Log-Mel representation applies logarithmic compression to filterbank energies. It is commonly used as a compact two-dimensional input in which one axis represents time and the other represents frequency bands. The project's primary configuration uses 40 Mel bins, a 25 ms analysis window, and a 10 ms hop at 16 kHz. The exact FFT size and implementation must be fixed in the experiment configuration and held constant across comparisons.

Mel-Frequency Cepstral Coefficients (MFCCs) are derived from log Mel-filterbank energies using a cosine transform to produce a compact set of coefficients. MFCCs have a long history in speech recognition (Davis & Mermelstein, 1980). Comparing MFCCs with Log-Mel features allows the project to test whether the feature representation affects command recognition under the same data splits and model settings. This comparison is an experiment to perform, not a result assumed beforehand.

Training Data Augmentation 2.4

Data augmentation creates altered training examples to expose a model to controlled variation. For speech commands, relevant transformations include mixing background noise at defined signal-to-noise ratios, applying small temporal shifts, modest speed perturbation, and limited gain variation. Transformations should preserve the command's identity and remain plausible for the intended use. Augmentation is applied to training data only in the primary evaluation protocol; validation and test sets remain unchanged so that evaluation reflects a fixed reference distribution.

SpecAugment applies masking operations to time and frequency regions of speech features and was introduced as a simple augmentation approach for speech recognition (Park et al., 2019). In this project, feature masking is an optional controlled experiment. Its effect should be evaluated on real held-out recordings, with each augmentation setting documented. Synthetic or augmented data should not replace real human-speech evaluation.

Neural Architectures for Small-Footprint KWS 2.5

Depthwise Separable Convolutional Neural Network 2.5.1 ((DS-CNN

A convolutional neural network learns local patterns from time–frequency features. Standard convolution mixes spatial filtering and channel combination in one operation. A depthwise separable convolution

separates these operations: a depthwise convolution filters each input channel independently, and a pointwise convolution combines channels. This factorization can reduce parameter count and multiply–accumulate operations, making DS-CNN a relevant baseline for resource-constrained keyword spotting (Zhang et al., 2017). The actual size and speed depend on the chosen layer .widths, kernels, strides, and implementation, so they must be measured

(Convolutional Recurrent Neural Network (CRNN 2.5.2

A CRNN combines convolutional layers with a recurrent layer. The convolutional part learns local time–frequency patterns, while a recurrent unit such as a Gated Recurrent Unit (GRU) or Long Short-Term Memory (LSTM) models temporal dependencies across frames. This structure can represent how acoustic evidence develops over the duration of a command. A bidirectional recurrent layer can use the full one-second segment in offline classification; a causal or unidirectional design is more appropriate when streaming latency is a central requirement. Performance on English benchmark data does not .guarantee the same ranking on Arabic data

(Keyword Transformer (KWT-1 2.5.3

A Transformer uses self-attention to model relationships between input positions. In keyword spotting, a compact Keyword Transformer can operate on patches or embeddings derived from time–frequency features and use attention to combine information across frames. Berg et al. (2021) describe the Keyword Transformer approach. KWT-1 is included as an optional higher-capacity comparison when computational resources permit. Its suitability for a target device should be determined by measured model size, memory use, and inference latency rather than assumed from recognition .accuracy alone

Bilingual Labels and Dataset Characteristics 2.6

The Arabic Speech Commands dataset is described as containing 12,000 WAV recordings, 40 Modern Standard Arabic command classes, and 30 speakers. Its documented split contains 7,200 training clips, 2,400 validation clips, and 2,400 test clips, with speakers kept separate across the three subsets. The limited speaker count and Modern Standard Arabic coverage constrain .(claims about dialectal generalization (Ghandoura et al., 2021

Google Speech Commands was introduced as a dataset for limited-vocabulary speech recognition (Warden, 2018). The project notes describe v0.02 as having approximately 105,829 files and 35 words, but the downloaded archive and version must be verified before training. The official

validation and testing lists should be preserved when available. The English corpus is larger and more diverse in speakers and recording conditions than the Arabic corpus, so a pooled bilingual score alone may conceal differences between languages.

The dataset analysis identifies 19 literal lexical matches: the digits zero through nine, yes/no, up/down/left/right, forward/backward, and stop. These matches form the primary shared-vocabulary experiment. Other Arabic or English labels remain language-specific. Similar meanings do not imply identical spoken labels: for example, Arabic words meaning “enable” and “disable” are not literal lexical matches for English “on” and “off”. A documented mapping table prevents accidental merging of different classes.

Unknown and Silence Rejection 2.7

A closed-set classifier always chooses one of its known classes unless the design explicitly provides a rejection mechanism. A command system should therefore include examples representing silence and out-of-vocabulary speech. Background-noise segments and non-target words can be used to construct negative examples, with sampling and class balance documented. The negative examples should be diverse enough to avoid teaching the model that a narrow set of background sounds represents all non-command audio.

At inference time, a confidence threshold can reject predictions whose maximum class probability is too low. The threshold should be selected using validation data and then fixed before final testing. In streaming systems, temporal smoothing or requiring agreement across consecutive frames can reduce unstable predictions; a cooldown interval can limit repeated actions. These mechanisms introduce a trade-off between missed commands, false triggers, and response delay.

Generalization and Experimental Validity 2.8

Generalization is the ability of a model to perform on examples that differ from the training data while still belonging to the intended task. For speech commands, speaker identity, accent, microphone, background noise, speaking rate, and recording environment can all affect the signal. If recordings from the same speaker appear in both training and test sets, a model may benefit from speaker-specific characteristics, making the test estimate overly optimistic.

The Arabic dataset provides a speaker-disjoint split, which should be preserved. For Google Speech Commands, the official split lists should be used when available. All model selection, augmentation tuning, and

confidence-threshold selection should use training and validation data only; the fixed test sets should be evaluated after these choices are complete. A separately recorded challenge set from new speakers would provide additional evidence of real-world generalization, if feasible. Corpus version, split .provenance, random seeds, and preprocessing settings should be documented

Evaluation Metrics 2.9

Accuracy is the proportion of all evaluated examples classified correctly.
It is easy to interpret but may hide weak performance on less frequent classes. Macro-F1 is the unweighted mean of the F1-score computed separately for each class, making it useful when class frequencies differ. Precision measures the fraction of predictions for a class that are correct; recall measures the fraction of true examples of a class that are detected. A confusion matrix shows which classes are mistaken for one another and can reveal confusable command .pairs

For negative-class handling, the false acceptance rate (FAR) measures
how often negative examples are incorrectly accepted as commands, while the false rejection rate (FRR) measures how often target commands are rejected or missed at the selected threshold. For continuous operation, false triggers per hour and detection latency are also useful. Efficiency should be described using model parameter count, serialized model size, runtime memory, computational cost, and inference latency on a specified device. These measures should be .reported separately by language and model where possible

Summary 2.10

This chapter introduced the theoretical foundations needed for the proposed system: the distinction between KWS and ASR, standardized audio processing, Log-Mel and MFCC features, data augmentation, compact neural architectures, bilingual label alignment, negative-class rejection, generalization, and evaluation metrics. Together, these concepts define the basis for a controlled comparison of Arabic-only, English-only, and bilingual command-recognition systems. All numerical outcomes remain to be .established through the planned experiments

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